

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Ylanage Linked Lists, Release Memory Correctly, Verify with Valgrind

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main() {
    int size = 2;
    int count = 0;
    int *numbers = malloc(size * sizeof(int));

    for (int i = 0; i < 5; i++) {
        if (count == size) {
            size = size * 2;
            numbers = realloc(numbers, size * sizeof(int));
        }

        numbers[count] = i + 1;
        count++;
    }

    printf("Numbers: ");

    for (int i = 0; i < count; i++) {
        printf("%d ", numbers[i]);
    }

    printf("\n");

    free(numbers);

    return 0;
}
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 2$ gcc task2.c -o task2
geya@LenovoL480:~/OS/Skill/Week 2$ ./task2
Numbers: 1 2 3 4 5
geya@LenovoL480:~/OS/Skill/Week 2$ |
```